Multi User Setup

BuMoChi

This is the canonical collaborative setup. It follows BuMoChi's **distributed sources, local synthesis** model: OSCGroups distributes route-free motion-source frames; every workstation's SuperCollider/BuMoChi process receives all sources, independently constructs the complete animation scene, and sends the finished local result to its own decoder and Godot renderer.

Operational model

Receiving and sharing animation data

Local animation sources

Data from local motion-capture sources such as XR-Animator or Waidayo is converted from VMC into route-free Bunraku frames by BunrakuOSCEncoder. The encoder sends an identical copy of every frame to two local destinations:

1. Bmc on port `57130`, for immediate local synthesis.
2. OscGroupClient on port `22244`, for distribution to remote workstations.

Remote animation sources

Each remote workstation performs the same two-way encoder fan-out. OSCGroups transports its route-free source frames to the local `OscGroupClient`, which forwards them to Bmc on port `57130`. Remote frames therefore enter through the same Bmc input as the encoder's direct local copy, while their stable `--avatar` and `--source` identities distinguish one motion source from another.

Animation data input

Both input paths converge on every workstation's Bmc at port `57130`:

1. Its own encoder's local source-frame copy. This is animation data from a local mocap system, such as XR-Animator with a webcam.
2. Remote source frames delivered by its `OscGroupClient`. These are animation data from remote mocap systems on computers connected through OSCGroups.

Animation data output

After combining, filtering, layering, and assigning motions to figures and avatars in SuperCollider, Bmc sends the completed animation frames to its local decoder on `39538`. Each frame is an OSC message containing the VMC destination port assigned to its avatar. The decoder reconstructs a VMC bundle, forwards it to the specified local Godot receiver, and Godot renders the avatar.

SuperCollider/Bmc never sends processed output to OSCGroups. The final animation is synthesized and rendered locally. Because every workstation receives the shared source frames and uses the same session/composition and Godot scene, each workstation independently constructs and renders the same intended animation.

This is analogous to networked sound synthesis in sc-hacks-redux: collaborators share source material, while every workstation performs the full synthesis locally.

One client per workstation

One full-duplex (input/output) `OscGroupClient` handles both network directions:

```
local encoder -> client input port 22244 -> OSCGroups network
OSCGroups network -> client output port 57130 -> local Bmc
```

Do not start separate sending and receiving clients on an ordinary workstation.

Shared-scene rule

All workstations should load the same Bmc session/composition and run the same Godot scene, including the same avatar-to-VMC-port assignments:

Source stream	Completed avatar	Godot character	VMC port
`PerformerA`	`Ishidomaru`	Ishidomaru	39539
`PerformerB`	`Mother`	Mother	39540

Both Bmc instances receive both source streams and synthesize both completed avatars locally. Both Godot instances therefore render the same defined scene. Network latency can cause small timing differences.

The OSCGroups server normally does not echo a workstation's own message back. This is expected: the encoder supplies its local Bmc copy directly.

Signal path on every workstation

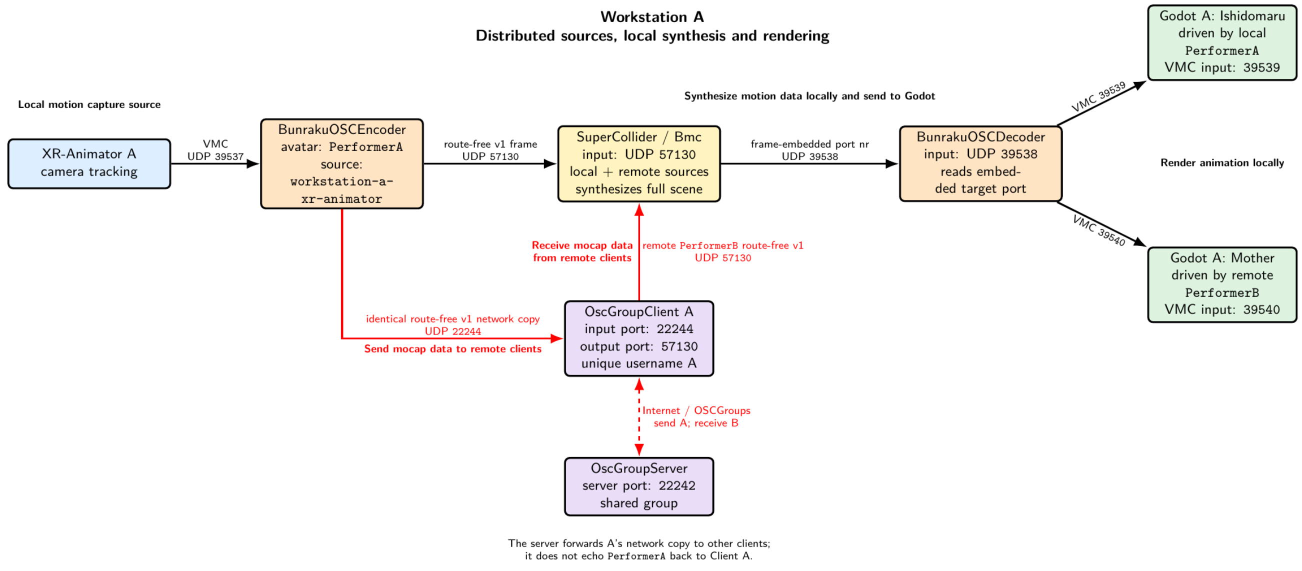
```
local XR-Animator
    -> VMC 127.0.0.1:39537
local BunrakuOSCEncoder
    -> route-free v1 127.0.0.1:57130 (local Bmc)
    -> identical route-free v1 127.0.0.1:22244 (OscGroupClient)
one local OscGroupClient
    -> OSCGroups server and remote collaborators
remote route-free v1 frames
    -> same local OscGroupClient
    -> 127.0.0.1:57130 (local Bmc)
local Bmc
    -> synthesizes the complete scene from local and remote sources
    -> routed v2 frames 127.0.0.1:39538 (local decoder only)
one local BunrakuOSCDecoder
    -> local Godot VMC ports 39539, 39540, ...
```

Only route-free source frames cross the network. Clips, motions, figures, avatar assignments, final routing, decoding, and rendering remain local.

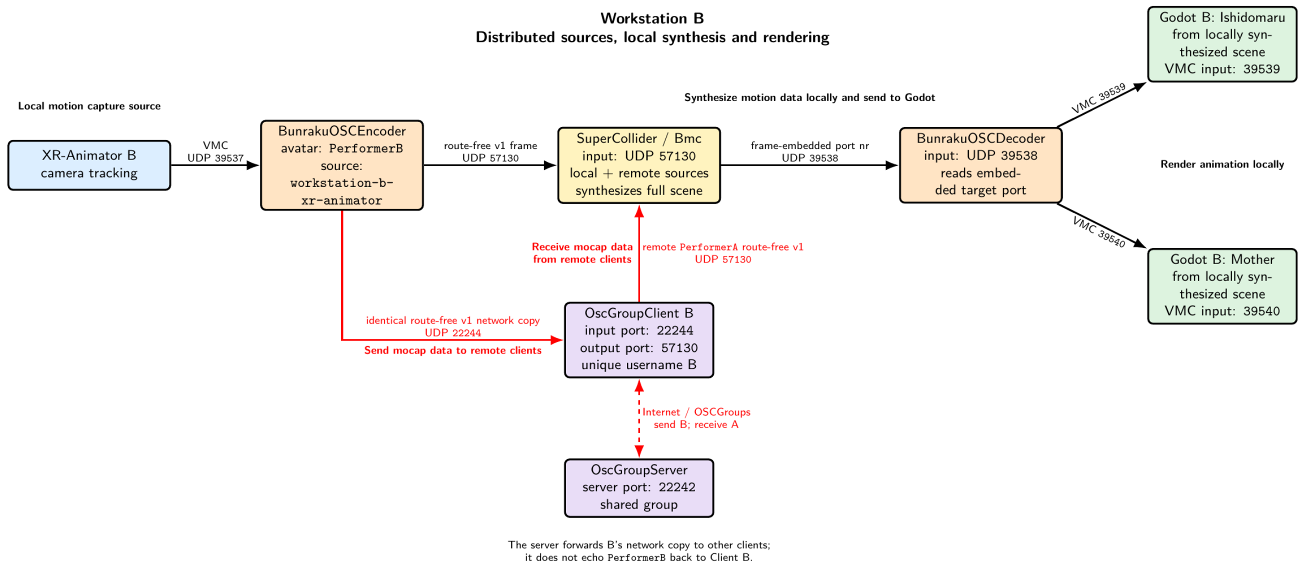
Complete workstation pipelines

The diagrams show two workstations. Both synthesize the same scene: Ishidomaru listens on `39539` and Mother on `39540`.

Workstation A



Workstation B



Shared and unique settings

All collaborators must agree on the OSCGroups server, group credentials, stable source identities, Bmc session/composition, Godot scene, and avatar-to-port map. Every workstation needs a unique OSCGroups username. Local application ports, including client input port `22244`, may use the same numbers on different computers because each port belongs to a different host.

Port map on each workstation

Port	Listener	Sender
39537	local BunrakuOSCEncoder	local XR-Animator
57130	local Bmc/SuperCollider	local encoder and `OscGroupClient` rx
22244	local `OscGroupClient` tx input	local encoder
39538	local shared decoder	local Bmc synthesized outputs
39539	first local Godot VMC receiver	local decoder

Port	Listener	Sender
39540	second local Godot VMC receiver	local decoder

Startup procedure

1. Start Godot

Run the agreed scene. In this example Ishidomaru listens on `39539` and Mother on `39540`.

2. Start the local decoder

```
BunrakuOSCDecoder \
  --listen-port 39538 \
  --allow-target-port 39539 \
  --allow-target-port 39540 \
  --verbose
```

The allow-list is optional.

3. Configure and start Bmc

Both workstations load the same session/composition. Minimal routing:

```
(
Bmc.reset;
Bmc.addAvatar(\Ishidomaru, "Ishidomaru");
Bmc.avatar(\Ishidomaru).vmcPort_(39539);
Bmc.addAvatar(\Mother, "Mother");
Bmc.avatar(\Mother).vmcPort_(39540);
Bmc.decoderPort_(39538);    // default
Bmc.forwardDecoder_(true); // false disables local renderer output
Bmc.start(57130);
)
```

`Bmc.reset` restores decoder port `39538` and enables decoder forwarding.

4. Start one OscGroupClient

```
HelperAppsAndExamples/OSCGroups/bin/macos/OscGroupClient \
  SERVER_ADDRESS 22242 LOCAL_TO_REMOTE_PORT 22244 57130 \
  USER_NAME USER_PASSWORD GROUP_NAME GROUP_PASSWORD
```

`22244` is the client's input port (`localTxPort` in the command-line interface), and `57130` is its output port (`localRxPort`). Remote route-free frames enter the same Bmc input as local frames.

5. Start the local encoder

Workstation A:

```
BunrakuOSCEncoder \
  --avatar "PerformerA" \
  --source "workstation-a-xr-animator" \
  --verbose
```

Workstation B uses `PerformerB` and `workstation-b-xr-animator`. Encoder defaults are local Bmc `57130` and local `OscGroupClient` `22244`.

Optional controls:

```
--bmc-ip 127.0.0.1 --bmc-port 57130
--oscgroups-ip 127.0.0.1 --oscgroups-port 22244
--no-bmc          # network-only diagnostic mode
--no-oscgroups    # local-only mode
```

At least one output must remain enabled.

6. Enable XR-Animator

Set its VMC destination to `127.0.0.1:39537`.

Verify

1. Each encoder's `received`, `sent`, `bmc_sent`, and `oscgroups_sent` counts increase.
2. Each Bmc runs on `57130` and receives both source identities.
3. Each client registers successfully.
4. Each local decoder receives only locally synthesized routed frames from Bmc on `39538`.
5. Ishidomaru moves on `39539` and Mother on `39540` in both Godot scenes.

Avoid feedback and identity errors

- Give each encoder a unique `--avatar` and `--source` pair.
- Send only encoder-produced route-free source frames to `22244`.
- Never send Bmc routed output, decoder output, or Godot VMC output to OSCGroups.
- Keep client `localTxPort 22244` and `localRxPort 57130` distinct.
- Never set `localRxPort` to `22244`, which creates a loop.
- Never set `localRxPort` to `39538`, which bypasses local synthesis.
- Do not expect the server to echo local frames; the encoder provides the local copy.

Diagnose ports

```
lsof -nP -iUDP:39537
lsof -nP -iUDP:57130
lsof -nP -iUDP:22244
lsof -nP -iUDP:39538
```

Shutdown

Disable XR-Animator, evaluate `Bmc.stop`, stop the encoder/client/decoder with `Control-C`, and stop Godot.

Related detail

- OscGroupClient Use and Configuration
- OSC Encoder and Decoder Use and Configuration
- Avatar Port Numbers
- Port Number Specification